

Beyond Smart Vehicles: Rethinking Intelligent Junctions for Bus Reliability and VRU Safety

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1 Introduction

A bus waits at a red signal even though no vehicles are approaching from the other direction. A pedestrian is still crossing the street when the signal changes and vehicles begin to move. These everyday moments reveal how urban mobility is often shaped not by advanced vehicles or digital platforms, but by the behaviour of signalised junctions.

The United Kingdom is entering a new phase of intelligent transport innovation. Autonomous vehicles, artificial intelligence and connected infrastructure promise to reshape how people and goods move through cities. Automated mobility services are already being prepared for deployment in London.

However, transport inefficiencies remain visible. As illustrated by the two examples above, buses can lose valuable minutes waiting for the next signal phase even when conflicting traffic demand is minimal—a scenario I have occasionally observed during early-morning travel to the airport. Pedestrian crossings are often timed according to assumed walking speeds, meaning that slower pedestrians may still be in the crossing when the signal phase changes. These examples highlight a broader challenge: while intelligence is advancing rapidly in vehicles and digital platforms, the streets where mobility is experienced remain comparatively less responsive.

In the U.K., the objectives of Transport 4.0 sit alongside commitments to bus reliability and the Vision Zero goal of eliminating serious road harm. Achieving these goals requires more than smart vehicles or centralised platforms. Junction infrastructure itself must become capable of coordinating buses, vulnerable road users (VRUs) and vehicles in real time.

This essay discusses the role of data, AI and digital infrastructure in enabling such junction intelligence. As these technologies continue to develop over the coming years, they have the potential to transform signalised junctions from passive control points into adaptive systems that actively coordinate traffic demands in real time. Drawing on my research in AI-based multimodal traffic signal control and field deployment experience in Germany, this essay argues that augmenting existing signal controllers with intelligence offers a practical pathway for improving bus priority and protecting VRUs across existing urban networks.

2 Why This Matters in the U.K. Now

Across the U.K., transport policy increasingly emphasises automated, connected and data-driven mobility. The government has indicated that self-driving vehicles could begin operating on British roads as early as 2026 following the introduction of new automated vehicle legislation (Centre for Connected and Autonomous Vehicles, 2025; Department for Transport, 2024b; U.K. Parliament, 2024). Companies such as the British start-up Wayve are developing embodied AI-based driving technologies in London, signalling rapid progress in automated mobility.

However, advances in vehicle intelligence do not eliminate the operational challenges that already exist in everyday urban traffic environments. Many of these challenges are experienced most clearly in public transport operations and pedestrian safety.

Improving bus reliability remains a central objective of transport policy, particularly in major cities where bus networks form the backbone of everyday mobility. The National Bus Strategy identifies bus priority and improved journey time reliability as key measures for strengthening the role of buses in the transport system (Department for Transport, 2021, 2024a). Delays at signalised intersections continue to affect service regularity and passenger experience.

Road safety policy similarly prioritises the protection of VRUs. London’s road safety framework calls for safer streets and improved junction design to reduce risks for pedestrians and cyclists (Transport for London, 2013). These principles build on a longer tradition of U.K. road safety policy promoting safer infrastructure and stronger protection for VRUs (Department for Transport, 2000). Comparative studies of national road safety strategies have examined how countries such as Sweden, the U.K., the Netherlands and Australia have developed system-oriented approaches to road safety. These approaches increasingly adopt Vision Zero principles, recognising human vulnerability and the inevitability of human error while seeking to design transport systems that prevent such errors from resulting in severe injury or death (Hughes et al., 2015; Mendoza et al., 2017).



Figure 1: The Southampton Row/Theobalds Road junction at Holborn is one of the most dangerous junctions in London (BBC News, 2022).

Many of these challenges converge at signalised junctions, where buses, pedestrians and other traffic streams interact within a shared space. Existing control systems such as the widely deployed SCOOT[®] already provide adaptive traffic management, yet many operational decisions still rely on predefined

parameters and assumptions about traffic demand or road user behaviour. Replacing these systems entirely would be costly and disruptive. A more practical pathway is to augment existing infrastructure with additional digital intelligence that allows junctions to respond more effectively to real-time conditions.

3 Bus Reliability: From Passive Priority to Context-Aware Junction Control

Buses remain one of the most accessible and flexible forms of public transport, carrying millions of passengers each day. Ensuring reliable bus services is therefore a central objective of transport policy. Measures such as bus lanes and traffic signal priority are widely used to improve bus journey times and reliability (Department for Transport, 2021, 2024a). Supporting guidance such as Local Transport Note 1/24 similarly identifies bus priority as an important mechanism for reducing delays and improving service consistency (Department for Transport, 2024a).

From a passenger perspective, service reliability strongly influences the attractiveness of bus transport. A study examining several bus routes in Birmingham found that passenger satisfaction is closely linked to perceived service quality and operational performance (Farayibi & Udechukwu, 2023). When bus services become unreliable, passengers are more likely to shift toward private car use.

Transit signal priority has long been used to support bus operations at signalised intersections. Traditional systems rely on vehicle detection or predefined priority requests when a bus approaches an intersection. While effective in some situations, these methods often operate in a relatively passive manner, and cannot fully account for the broader traffic conditions at the junction.

Evidence from British cities illustrates how intersections influence bus reliability. An analysis of operational data from Birmingham’s Route 63 found that the number of signalised intersections and pedestrian crossings significantly contribute to bus travel time delays (Kaewunruen et al., 2021). Interactions between buses, traffic signals and pedestrian crossings can therefore become a major source of service variability.

These limitations highlight the need for more context-aware junction control. By interpreting real-time traffic conditions—such as approaching buses, traffic queues and VRU demand—AI-augmented junction intelligence can move beyond simple priority requests toward more adaptive signal control that improves bus reliability while maintaining safe operations.

4 VRU Protection: From Fixed Clearance to Adaptive Protection

Protecting VRUs has also become a central concern in transport policy. Pedestrians and cyclists face an unacceptable risk of serious injury, particularly at intersections where multiple traffic streams intersect. Vision Zero strategies adopted in many British cities aim to address this challenge by recognising that human error is inevitable and by designing streets so that mistakes are less likely to result in severe harm (Mendoza et al., 2017). In London, the Safe Streets for London programme reflects this approach through intersection improvements, lower speed limits and targeted safety measures (Transport for London, 2013).

Signalised crossings play a key role in supporting safer streets. However, many signal timings still rely

on fixed assumptions and settings about pedestrian walking speeds and clearance times. These assumptions and settings cannot always reflect the diversity of people using the street. Older pedestrians or individuals with reduced mobility may require more time to cross safely.

Research in Greater Manchester found that concerns about crossing safety were among the most significant barriers affecting the mobility of older residents (Musselwhite, 2021). Participants frequently identified inadequate crossings and unsafe street environments as major factors limiting their mobility and independence.

More intelligent junctions offer an opportunity to address these challenges. By detecting the presence and movement of VRUs in real time, intelligent junctions can adapt signal timings dynamically and ensure that crossings remain safe for a wide range of road users.

5 Why Intelligence Belongs at the Junction

If infrastructure is to respond effectively to complex urban interactions, intelligence must be located close to where these interactions occur: the junction. While cloud-based analytics and connected vehicle systems offer powerful tools for network-level optimisation, many operational decisions at signalised intersections must be made within seconds.

Latency, connectivity constraints and data governance considerations limit the practicality of relying solely on centralised computing systems. Locating intelligence directly at the junction therefore allows traffic signals to respond to real-time road user behaviour and demand. By combining sensing, data processing and AI-based decision support at the roadside, intelligent junction infrastructure can interpret local traffic conditions and adjust signal operations accordingly.

6 Lessons from a Field Deployment in Germany

The next stage of intelligent transport should not depend solely on smarter vehicles, but also on more intelligent infrastructure. Many existing traffic signal systems already include adaptive logic, yet important operational processes still depend on fixed parameters such as gap-out values, priority rules or assumed pedestrian timings. A pilot Edge-AI deployment in Germany explored how junction intelligence could make intersections more responsive without replacing the existing controller.

As part of my Ph.D. research, a field deployment was carried out at an actuated signalised intersection in Leipzig, Germany. The objective was not to replace the existing traffic control system, but to augment it with Edge-AI-based support, as shown in Fig. 2. Existing detectors and pedestrian buttons continued to provide conventional vehicle detection and demand inputs, while a LiDAR sensor mounted on a traffic signal pole detected pedestrian and cyclist activity. These data were processed locally through edge computing, allowing the junction to interpret real-time traffic conditions while meeting cybersecurity requirements.

This intelligence addressed both challenges discussed earlier. For bus operations, the system used an AI-based bus approaching model to estimate bus arrival times and generate priority requests. AI-based queue estimation enabled more flexible phase extension decisions beyond fixed gap-out parameters, allowing the junction to respond more effectively to real-time traffic conditions.

For VRUs, LiDAR detection triggered virtual pedestrian demand signals when pedestrians were wait-

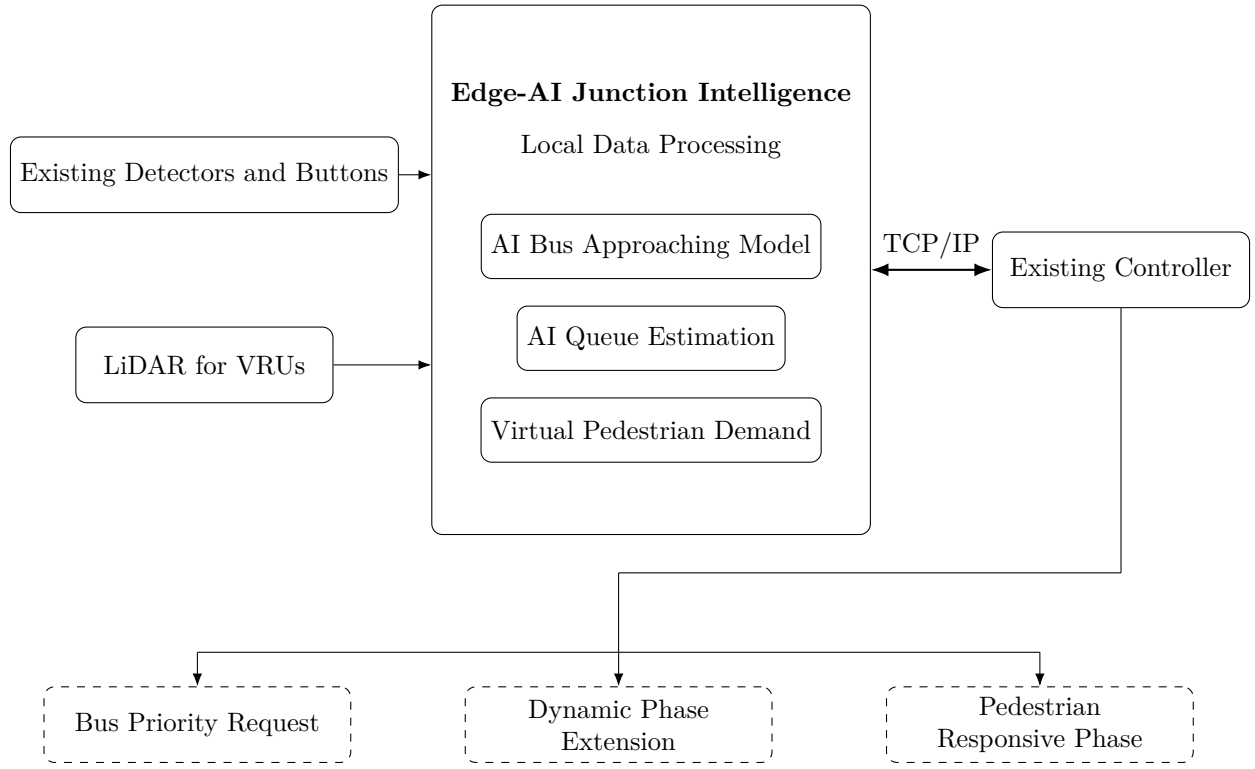


Figure 2: Field deployment architecture. Existing detectors, pedestrian buttons and LiDAR sensing were integrated through an Edge-AI layer operating alongside the existing signal controller, enabling bus priority requests, dynamic phase extension and pedestrian-responsive control.

ing to cross, enabling faster and more responsive signal changes than conventional fixed demand logic. Both augmentations were deployed on an edge computing device, and communicated with the existing signal controller via a TCP/IP interface.

The deployment operated continuously for three months, and demonstrated that intelligence can operate within an existing control system. Its significance lies not only in the technology itself, but in what it enables: more reliable bus movement, more responsive protection for vulnerable road users, and a practical pathway for modernising existing intersections without wholesale infrastructure replacement.

7 Conclusion

Data, AI and digital technologies are already transforming how transport networks operate. While much attention has focused on intelligent vehicles, this essay argues that the next stage of smart mobility must also include intelligent junctions. Signalised junctions represent a critical point where operational inefficiencies and safety risks often emerge.

By integrating sensing technologies, real-time data and AI-based tools at the roadside, junctions can evolve from passive signal controllers into intelligent nodes within the wider transport network. Such systems can interpret real-time traffic conditions, improve bus reliability, and provide more responsive protection for VRUs. Moreover, smarter infrastructure can help maximise the potential benefits of emerging technologies such as automated vehicles by providing more predictable and adaptive traffic environments. Augmenting current junctions with Edge-AI-based intelligence offers a realistic and scalable pathway for cities to modernise traffic control without replacing existing infrastructure.

As emerging technologies continue to develop over the coming years, the future transport network will increasingly depend on cooperation between intelligent vehicles and intelligent streets. Delivering a truly smart transport network therefore requires moving beyond the smart vehicle alone, and rethinking how junction intelligence can support reliable and safe mobility.

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